

National-scale mapping of staple crops in Nepal with multi-year Sentinel-2 L2A imagery and binary transformer models

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ABSTRACT

Accurate mapping of staple crops using satellite imagery is essential for sustainable agricultural management, particularly in data-limited regions. Despite recent advances in Earth observations and deep learning, using multi-year time series satellite data and binary deep learning models to produce national-scale multi-class crop type map remains underexplored. In this study, we attempt to map staple crops, other crops, and non-crop at 10 m resolution that leverages multi-year, unsmoothed Sentinel-2A/B MSI Level-2A time series imagery, binary transformer models, and a probability-based thresholding method. Nepal was selected as the case study due to its complex topography and heterogeneous landscapes. We developed six experimental setups, and models achieved an average overall accuracy of 91.39%, with class-wise F1-scores ranging from 0.81 to 0.96. The best-performing model was then applied to 28 tiles covering Nepal, and a probability-based thresholding approach was used to assemble binary results into a multi-class map. The map overall accuracy was higher than 94%, when validated with field data. Sensitivity analyses further confirmed the robustness and spatial consistency of the proposed framework across diverse land-cover and agroecological zones. Finally, the produced national-scale maps indicated 3.91% increase in staple crops areas from 2021 to 2023 in Nepal.

1. Introduction

Climate change is intensifying food insecurity, specifically in developing countries where smallholder farmers are highly vulnerable to climate variability (West et al., 2019). In response, Climate Smart Agriculture (CSA) (Climate-smart agriculture sourcebook, 2014) has emerged as a key paradigm aimed at enhancing agricultural productivity, strengthening resilience to environmental change, and reducing greenhouse gas emissions. To evaluate CSA outcomes at a national scale, the availability of accurate, customized, and spatially explicit crop type maps at spatial resolutions appropriate for smallholder farming systems is essential. This need is especially critical for staple crops (e.g., paddy rice, maize, wheat, and potatoes), which are central to local diets and nutritional security. Despite advances in satellite data, many existing crop-type maps remain inadequate for CSA applications in smallholder systems due to limitations in spatial resolution, geographic coverage, or both. For example, the USDA Crop Data Layer provides 30-meter maps for the United States (Boryan et al., 2011), while other regional products

typically range from 10 to 30 m (Blickensdörfer et al., 2022; d'An-drimont et al., 2021; Li et al., 2023; Orynbaiyzy et al., 2019). However, these national scale crop maps are available for only a limited number of countries, and many developing countries still lack this fundamental dataset.

For national-scale staple crop classification at 10 m spatial resolution, Sentinel-2 MultiSpectral Instrument (MSI) data provides a highly suitable and cost-effective data source. The twin satellites, Sentinel-2A and 2B, launched in 2015 and 2017 respectively, offer near-global coverage with a revisit frequency of approximately five days and have generated nearly a decade of consistent imagery (Segarra et al., 2020). Vegetated surfaces such as crops, grasslands, and forests exhibit distinct seasonal changes in spectral reflectance driven by phenological development. These temporal patterns, captured through vegetation indices (Gao and Zhang, 2021; Zhang et al., 2013), enable discrimination vegetated from non-vegetated land covers. Sentinel-2 MSI data have therefore been widely adopted for vegetation-related classification challenges (Belgiu and Csillik, 2018; Harkort et al., 2025; Lambert et al.,

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2018; Xie et al., 2025). In addition, the red-edge bands of Sentinel-2 are particularly sensitive to vegetation's structural and biochemical properties and have shown significantly improvement on crop type differentiation (Dong et al., 2019; Guerini Filho et al., 2020). Currently, the European Space Agency (ESA) and Google Earth Engine (GEE) are two major platforms for data accessibility. To avoid data pre-processing difficulty, studies have used GEE for feature extraction across spatial scales (Brovelli et al., 2020; Liu et al., 2020).

A growing body of research highlights the importance of time series satellite data for crop type mapping. Approaches based on multi-scale label hierarchies (Turkoglu et al., 2021), time-weighted dynamic time warping (Belgiu and Csillik, 2018), and phenology-driven classification (Liu et al., 2020; Nguyen et al., 2020) demonstrate that temporal information is critical for distinguishing crop types. This is because crop phenology, as captured by multi-temporal satellite observations, provides distinctive temporal signatures that support crop type separation (Veloso et al., 2017; Zeng et al., 2020). However, satellite time series data are frequently affected by cloud and cloud shadow contamination (Qiu et al., 2019; Zhu and Woodcock, 2012). As a result, these contaminated pixels will affect time series data quality. Thus, when considering time series satellite dataset, the major trade-off is creating synthetic observations through gap-filling (i.e., reducing influence from contaminated pixels using mathematical methods) or retaining available observations (i.e., just removing contaminated pixels and keeping the day-of-year date order). Common gap-filling strategies include compositing (Azzari and Lobell, 2017; Li et al., 2023), multi-sensor data fusion (e.g., combining all available sensor data), and percentile-based approaches (Sisheber et al., 2022). While effective at increasing temporal continuity, these approaches rely on synthetic or interpolated observations that may alter the original spectral-temporal structure, potentially introducing bias and degrading classification performance (Che et al., 2024; Zhao et al., 2021). An alternative is to retain original available observations while removing only cloud- and shadow-contaminated pixels, hereafter referred to as an unsmooth time series. This approach preserves the integrity of available observations but has received limited attention in crop type classification studies.

In recent years, deep learning has emerged as a leading methodology due to its flexibility and strong classification performance (Adrian et al., 2021; Kattenborn et al., 2021; Wang et al., 2022). Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) have been widely applied to extract spatial and temporal features from satellite imagery (Attri et al., 2023; Turkoglu et al., 2021). More recently, transformer-based models have gained prominence by leveraging attention mechanisms to dynamically weight informative portions of input data. Originally developed for natural language processing (Vaswani et al., 2017), transformers have been adapted for computer vision by converting imagery into one-dimensional sequences and incorporating positional encodings (Aleissae et al., 2023; Xu et al., 2024). Transformers offer several advantages for satellite time series analysis: (1) multi-head self-attention enables focused learning of temporally relevant patterns while feed-forward networks capture complex non-linear relationships, which is distinct from the hierarchical feature extraction used in CNNs or RNNs (Luo et al., 2023; Yao et al., 2023); (2) long and noisy time series can be processed without significant degradation performance; and (3) parallel processing and positional encoding improve computational efficiency (Han et al., 2023). Given the challenges posed by unsmooth time series data and the critical role of phenological variability in distinguishing crop types, remote sensing studies typically use encoder-only structure to better align with the characteristics of satellite data (Schiller et al., 2024; Zhang et al., 2024a). Importantly, since the unsmooth time-series data removes contaminated pixels, the length of the data can vary across samples. Due to the fact that the deep learning models can only process well-organized data, the original unsmooth data cannot directly be used as model input. Inspired from the padding method used in deep learning models and mask mechanism from transformer models, we padded the original

unsmooth data with a prespecified temporal length but still keeping phenology information. While encoder-only model training, the mask mechanism can allow the model to only focus on available observations in padded time series data. Despite its promise, there have been few applications of this approach for primarily rural national-scale staple crop mapping incorporating vegetation phenology with limited reference data.

As a multi-class task with satellite imagery, developing one model with sufficient reference data and discriminative classes can achieve decent results (Thorp and Drajat, 2021; Turkoglu et al., 2021). However, when facing issues such as class imbalance, class similarity and limited reference data, the use of one universal model may not achieve the desired performance. Thus, multi-class problems are often decomposed into a set of binary or class-specific models, such as one-vs-rest, one-vs-one, or hierarchical classifiers. For instance, Li et al. (2023) used probability layers generated from a random forest model to assign a pixel to maize or soybean based on the highest probability. Zhao and Qin (2022) used binary classification approaches (i.e., mangroves and non-mangroves) to identify mangroves in China by comparing with multi-class classification (i.e., mangrove and multiple non-mangrove classes), and their experimental results showed improved accuracy. However, there is still a knowledge gap in existing literature related to assembling a multi-class staple crop map with binary deep learning model results with a probability thresholding method.

Thus, the study aims to map staple crops at a national scale in a data-constrained, developing country using multi-year unsmooth Sentinel-2 L2A time series imagery and binary transformer models. Specifically, we: (1) employ transformer models optimized for unsmooth Sentinel-2 time series; (2) evaluate the effectiveness of phenology information for national-scale crop type classification; (3) assess model performance on multi-year time series dataset; and (4) introduce an innovative thresholding strategy to translate binary model outputs into a final multi-class result. Method robustness was assessed through multiple experimental configurations, while model sensitivity was examined through visual inspections across diverse agroecological zones.

2. Study area and dataset

2.1. Study area

This study was conducted in Nepal due to its predominance of smallholder farming systems, complex topography, high agroecological diversity, and limited availability of reference data. Nepal is located between latitudes 26°22'–30°27'N and longitudes 80°04'–88°12'E and covers approximately 147,181 km², with more than 80% of landmass situated in mountainous terrain (Joshi et al., 2021; Nepal et al., 2020). Due to the sharp elevation gradient from south to north, the country is divided into three agroecological zones (Fig. 1). The southern lowland Terai zone contains ~ 20% of the country's agricultural land and is dominated by staple crops such as paddy rice, wheat, and vegetables. The central Hill zone covers about 41.7% of the national territory, characterized by heterogeneous landscapes and mixed cropping systems, where maize is commonly intercropped with rice, wheat and vegetables. The Mountain zone, located in the northern high-altitude region, is largely covered by snow and ice; however, during warmer months, crops such as potato, maize, and buckwheat are cultivated in suitable areas. Maintaining the production of staple crops (i.e., paddy rice, maize, potato, and wheat) that supply a major proportion of dietary energy and nutrients is therefore a national priority (Khatri-Chhetri et al., 2017).

2.2. Satellite imagery

We used Sentinel-2 MSI L2A surface reflectance data, which comprises 13 spectral bands spanning wavelengths from 432.7 nm to 2286.5 nm. The L2A product is atmospherically corrected to mitigate the effects of aerosols, water vapor, and cloud contamination (Jérôme Louis et al.,

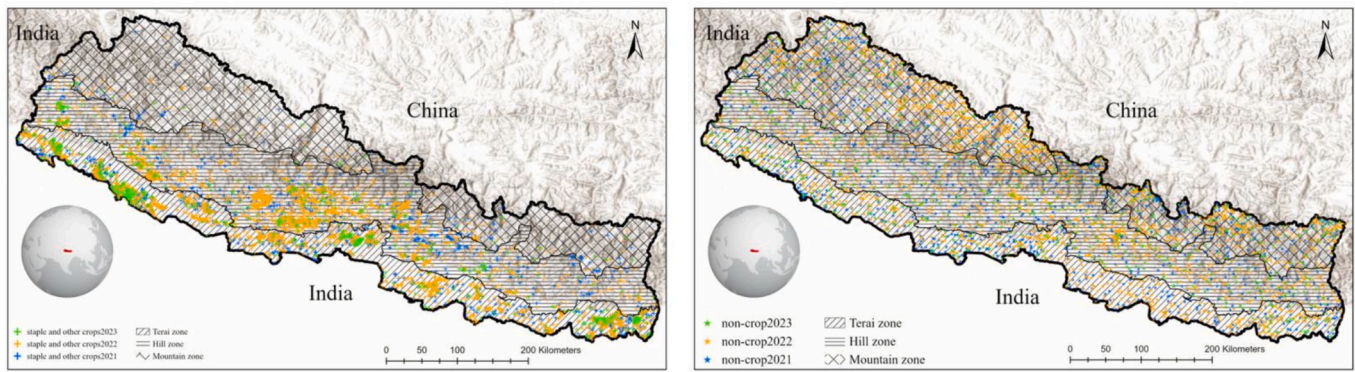


Fig. 1. Nepal country boundary, spatial distribution of crops (staple and other) reference data (left), and other land cover reference data (right). Reference points are categorized by land cover type and year of collection: crop data are represented by ‘+’ symbols, while other land cover types are indicated by stars. Color coding denotes the year of data collection: blue for 2021, orange for 2022, and green for 2023.

2016; Main-Knorn et al., 2017). Specifically, we selected blue, green, red, red-edge1, red-edge2, red-edge3, NIR (wide), NIR(B8A), shortwave infrared-1 (SWIR-1), and shortwave infrared-2 (SWIR-2) bands. This selection also enables the use of unique red edge bands, which have been shown to be particularly effective for vegetation monitoring (Choi et al., 2025; Delegado et al., 2011; Sun et al., 2020). All 20 m spatial resolution bands were resampled to 10 m using the nearest neighbor method (Luo et al., 2025; Xie et al., 2025).

A total of 28 Sentinel-2 tiles cover the study area (Fig. 3). To balance data availability with computational feasibility for deep learning model training, we applied a maximum cloud cover threshold of 30%, following metadata guidelines and established practices in the literature (Hunt et al., 2019; Liu et al., 2020). Under this threshold, the average number of images across all 28 tiles was 41.71, 61.96, 66.11 for the years 2021, 2022, and 2023, respectively (Fig. 2). To ensure consistency between platforms, we used the same acquisition dates identified in GEE to download corresponding Sentinel-2 L2A images (baseline N05) from the ESA platform (hereafter referred to as ESA data). This step was necessary to support national-scale staple crop mapping, given GEE’s limitation in handling computationally intensive trained deep learning models. For both GEE and ESA datasets, we utilized the Scene Classification Layer (SCL) to mask contaminated pixels (i.e., no-data, cloud shadow, medium- and high-probability cloud, and thin cirrus pixels). The number of available observations per tile in the GEE dataset ranged

from 2.04 to 41.25 in 2021, 9.86 to 61.36 in 2022, and 9.86 to 65.54 in 2023 at 10 m spatial resolution (Fig. 3).

3. Reference data

In this study, reference data were primarily sourced from the GeoKrishi mobile application, a digital platform jointly developed by Heifer International Nepal and Pathway Technologies. To improve accessibility for farmers without smartphones, the initiative also established digital resource hubs, referred to as *E-Chautaris*. As of December 2022, the GeoKrishi application had approximately 78,500 registered users of whom 57,554 contributed records included corrected geographic coordinates and crop type information.

While smartphone-based data collection offers a scalable solution for *in-situ* crop monitoring (Soler et al., 2024; Zhou et al., 2013), user-generated inputs can introduce spatial and thematic biases, such as field-edge recording, crop mislabeling, or clustering in densely populated areas. To enhance data reliability, we implemented a strict filtering procedure. All points were overlaid with the ESRI Sentinel-2 10-meter Global Land Cover Product from 2021 to 2023 (<https://gee-community-catalog.org/projects/S2TSLULC/>), which includes nine land cover classes (i.e., water, trees, flooded vegetation, crops, built area, bare ground, snow/ice, clouds, and rangeland) and achieves approximately 85% overall accuracy across classes (Karra et al., 2021). Points falling on

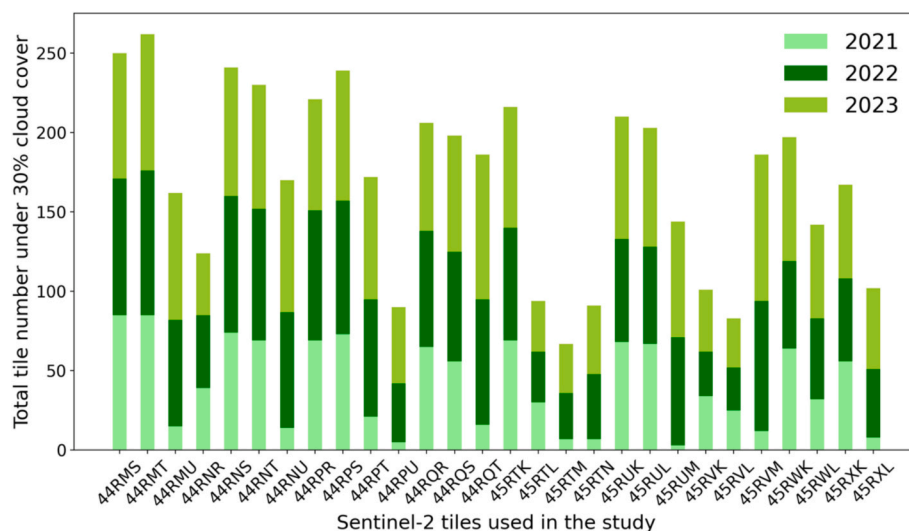


Fig. 2. Total number of Sentinel-2 L2A time series images per tile with cloud cover $\leq 30\%$ from Google Earth Engine (GEE). Bars are stacked by year: the bottom segment represents 2021, the middle and the top are 2022 and 2023, respectively.

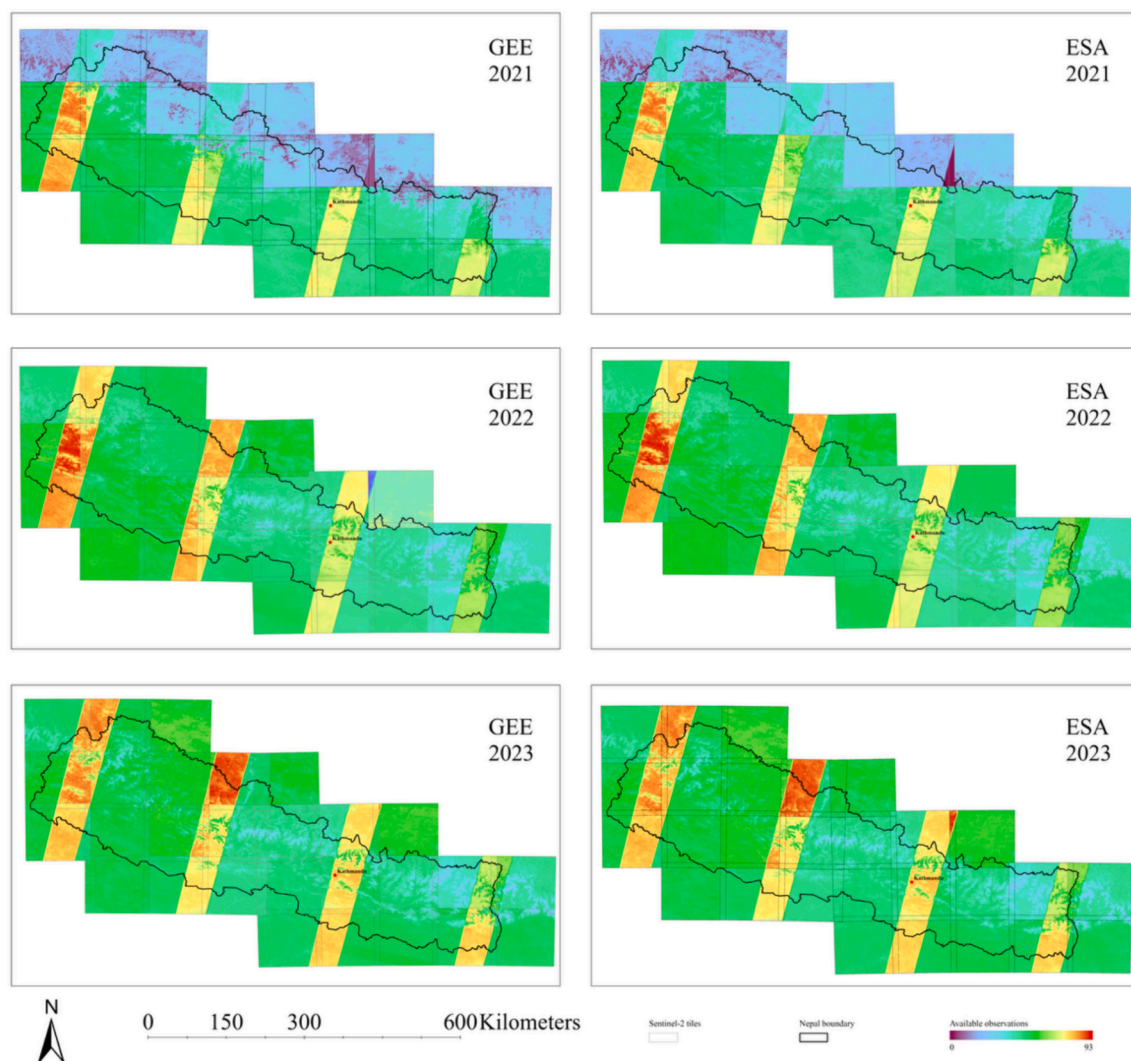


Fig. 3. Spatial distribution of the total number of available observations per pixel per tile at 10 m spatial resolution for the years 2021, 2022, and 2023, after removed cloud and cloud shadow contaminated pixels. Panels on the left represent imagery processed via GEE platform, while panels on the right correspond to data retrieved from the ESA platform.

non-crop pixels in any of three years were discarded, resulting in 13,204 high-quality reference crop points: 2,458 for 2021, 9,154 for 2022, and 1,592 for 2023 (Table 1).

To balance the reference dataset, we manually collected training points across six categories: trees, water, built-up areas, roads, bare ground, and snow/ice. Sample locations were identified using high-resolution imagery from the ESRI basemap (ArcGIS Pro v3.4.0). Spatial separation between points was enforced to reduce redundancy and ensure representative coverage across all provinces. This process yielded 1,119 points in 2021, 2,440 points in 2022, and 603 points in

2023 (Table 1). By integrating these additional points with the GeoKrishi crop data, we defined the standardized class in Table 2. Additionally, our team conducted field surveys across selected households during the 2023 harvest season. In total, 667 crop fields were surveyed nationwide, corresponding to 34,163 pixels at 10 m spatial resolution, including 298 staple crop fields and 369 other crop fields.

Table 1

Number of points for each class collected from GeoKrishi mobile application and authors.

Class name	Crop name	2021	2022	2023
Staple crops	Rice/Paddy	400	1487	60
	Potato	405	1222	152
	Maize	158	851	312
	Wheat	176	580	136
	sum	1139	4140	660
Other crops	--	1319	5014	932
Non-crop	--	1119	2440	603
sum	--	3577	11,594	2195

Table 2

Class definition and explanation used in the study.

Class name	Description	Explanation
Staple crops	Rice, potato, maize, wheat	Staple crop was defined based on the data provided from GeoKrishi mobile application.
Other crops	Other crops that smartphone data have	Other crops were defined based on the dataset from the mobile application.
Non-crop	Water body, building, road, bare ground, tree, shrub, snow, ice	Choose river, lake, building, road, bare ground, trees, shrubs, and snow/ice.
Either	any of above class	We designed this class based on threshold method.

4. Methods

The methodology of this study was carried out in a series of interconnected steps (Fig. 4).

4.1. Input data preparation

We utilized unsmooth Sentinel-2 time series data from 2021 to 2023. Each reference point was structured as a row, with spectral reflectance values arranged chronologically across available days of the year as columns. The maximum sequence length (T) was set to 93, corresponding to the largest number of available observations for any pixel across the three years. For points less than 93 available observations, we filled with -9999 for the rest of the sequence, and the value was consistent with established practices in previous study (Zhang et al., 2024a). The resulting dataset was formatted as a three-dimensional tensor with dimensions based on (number of reference samples, 93, number of spectral bands). We allocated 80% of the data for model training, and the remaining 20% for model validation.

4.2. Deep learning model explanation

The masked transformer model consisted of embedding and positional encoding layers, two masked transformer blocks, one average layer and one dense layer.

The prepared time series input $X = \{x_t \in \mathbb{R}^d | x_t \in \mathbb{R}^d | t = 1, 2, \dots, T\}$ and binary mask $M = \{m_t \in \{0, 1\} | t = 1, 2, \dots, T\}$ that worked at the end of the time series with -9999 as the filled value, where d was the 10 bands and temporal length T was 93. m_t was 1 if x_t is valid, otherwise $m_t = 0$. Then, the embedded input sequence was $Z_t = W_e x_t + P_t$, where $W_e \in \mathbb{R}^{d \times d_{dim}}$ was the embedding matrix, d_{dim} was embedding dimension, and P_t was the positional encoding. We complied two positional encoding strategies: (1) a dummy sequence representing chronological order and (2) incorporation of day-of-year (DOY) information.

The masked multi-head self-attention block included a masked multi-head self-attention layer, residual connections, layer normalization, and a point-wise feed-forward network with ReLU activation function (Glorot et al., 2011). We used multi-head attention layer from TensorFlow with attention_mask as one of the call arguments and set heads = 4. The output (T, d_{dim}) then had a dropout layer added to it to prevent model overfitting and was followed by a normalization layer. Then this output was followed by a pixel-wise feed-forward network with dropout and normalization layer.

The temporal length of features (T, d_{dim}) from the last masked multi-head self-attention block were used in a masked average pooling layer to generate single feature ($1, d_{dim}$). This step ensured only valid observations contribute to the final feature vector. The final prediction was $Y = \sigma(W_o z_{ag} + b_o)$, where $\sigma(\cdot)$ was an activation function, W_o and b_o were

learnable weight matrices and bias vector, z_{ag} was output from the masked average pooling layer.

To reduce model overfitting, we added L2 regularization on the multi-head self-attention layer and output layer to introduce a penalty on large weight values, thus, encouraging smoother model performance (Srivastava et al., 2014).

Due to the natural of the limited reference data (Fig.A1) and less than desired model performance from the three-class model (i.e., overall accuracy less than 63%), when distinguishing between staple and other crops classes, we created three binary models (Table 3). Hyper-parameters were rigorously optimized (Table 4), and sparse categorical cross entropy loss function with AdamW optimizer were applied. The use of CosineDecay warm-up learning rates (Zhang et al., 2024a) further mitigated overfitting during the model training. Each 2-class model was trained three times to account for randomness. Model implementation used Keras within TensorFlow, trained on $8 \times$ GPU Nvidia A30 with 24 GB memory each.

4.3. Probability threshold method to assemble final map

Using the fine-tuned parameters, we retrained the model with 100% training data to produce a generalized model. For each model, we produced three 2-class outputs: $A_{all \text{ crops-non}}$, $B_{other-non}$, and $C_{staple-non}$. In parallel, we produced probability layers: $AP_{all \text{ crops-non}}$, $BP_{other-non}$, and $CP_{staple-non}$.

We defined a pixel as a non-crop pixel if it satisfied the following condition:

$$\text{non-crop} = (A_{all \text{ crops-non}} == \text{non-crop}) \text{ or } (B_{other-non} == \text{non-crop}) \text{ or } (C_{staple-non} == \text{non-crop}) \quad (1)$$

To assign crop pixels as either staple or other crops, we applied an innovative thresholding method based on available field-survey data (Fig. 5). First, we evaluated a range of probability thresholds for classification within polygon areas of each tile from $BP_{other-non}$ and $CP_{staple-non}$ outputs. Specifically, threshold values ranging from 0.89 to 1.0 were tested for other crops and 0.75 to 0.86 for staple crops. These ranges were defined based on extensive experiments. Second, we calculated overall accuracy, producer accuracy, and user accuracy for both staple and other crops, and any threshold combination yielding an overall accuracy below 70% per tile was discarded. This step substantially

Table 3

Three binary models that used in the study.

Model name	Class name
Model A	all crops (staple and other) vs. non-crop
Model B	other crops vs. non-crop
Model C	staple crops vs. non-crop

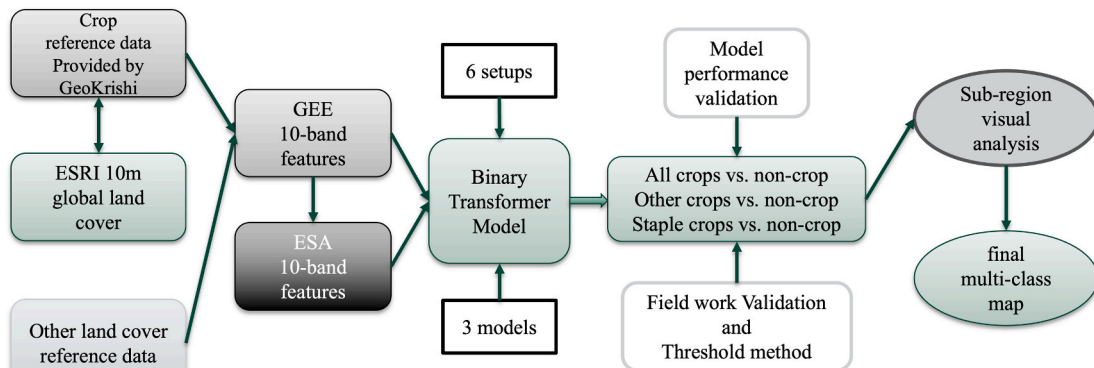


Fig. 4. The workflow of the study.

Table 4

Hyper-parameters needed for the model and all possible values we tested in the study. “Test values” column lists all possible values we tested. “Optimal values” column lists the best value.

Parameter name	Test values	Optimal values
Embedding dimension	32, 64, 128	64
Multi-head	2, 4, 8	4
Transformer blocks	1,2,3,4	2
Learning rate	0.01, 0.001, 0.0001	0.001
L2	0, 1e-2, 1e-3, 1e-4	1e-3
Dropout	0, 0.1, 0.2	0
Batch size	16, 32, 64, 128	32
Epoch	80, 100, 120	100

narrowed the feasible threshold ranges for both crop classes. Third, the refined threshold was applied to generate the final classification maps for staple crops and other crops. The resulting maps were then used to calculate overall accuracy, produce, and user accuracies from the summarized confusion matrix. Pixels that did not meet either threshold criterion were assigned to a fourth class (Table 2).

4.4. Validation methods

To comprehensively assess model robustness under varying conditions, we designed six experimental setups (Table 5). Model performance was validated with the independent 20% of reference data, following the widely accepted confusion-matrix-based method (Schiller et al., 2024; Zhao et al., 2022). We derived overall accuracy, producer and user accuracies, and the F1-score. Additionally, given Nepal ranges from flat southern plains to high northern mountains, we randomly selected one subregion (i.e., 1/16 of a Sentinel-2 tile, $2,745 \times 2,745$ pixels) in each agroecological zone to test model sensitivity. The selected subregions (Fig. 5) included: the Terai zone (44RNS-09) dominated by cropland; the Hill zone (45RTM-08) mixed land cover types, and the Mountain zone (45RWL-07) dominated by snow cover.

$$OA = \frac{TP + TN}{TP + TN + FP + FN} \quad (2)$$

$$UA = \frac{TP}{TP + FP} \quad (3)$$

$$PA = \frac{TP}{TP + FN} \quad (4)$$

$$F1score = \frac{2 \times UA \times PA}{UA + PA} = \frac{2TP}{(2TP + FP + FN)} \quad (5)$$

Where: OA was overall accuracy, UA was user accuracy, PA was producer accuracy. TP, TN, FP, and FN represented true positive, true negative, false positive and false negative from the confusion matrix.

5. Result

5.1. Model performance

See Fig. 6 reported model overall accuracy and most setups exceeded

Table 5

Six experimental setups and its explanation.

Name	Model component	Explanation
Setup 1	all 10 bands in 2022 extracted from GEE platform with dummy positional encoding	Null model
Setup 2	all 10 bands in 2022 extracted from GEE platform with DOY as positional encoding	Compare different positional encoding methods between setup1 and 2
Setup 3	all 10 bands in 2022 extracted from ESA platform with DOY as positional encoding	Compare different platforms between setup 2 and 3
Setup 4	7 bands (without red edge bands) in 2022 extracted from GEE platform with DOY as positional encoding	Test influence of red edge bands
Setup 5	all 10 bands in 2022 and 2023 extracted from GEE platform with DOY as positional encoding	Compare model performance between single year and two-year reference data
Setup 6	all 10 bands from 2021 to 2023 extracted from GEE platform with DOY as positional encoding	Test model performance on all three-year reference data

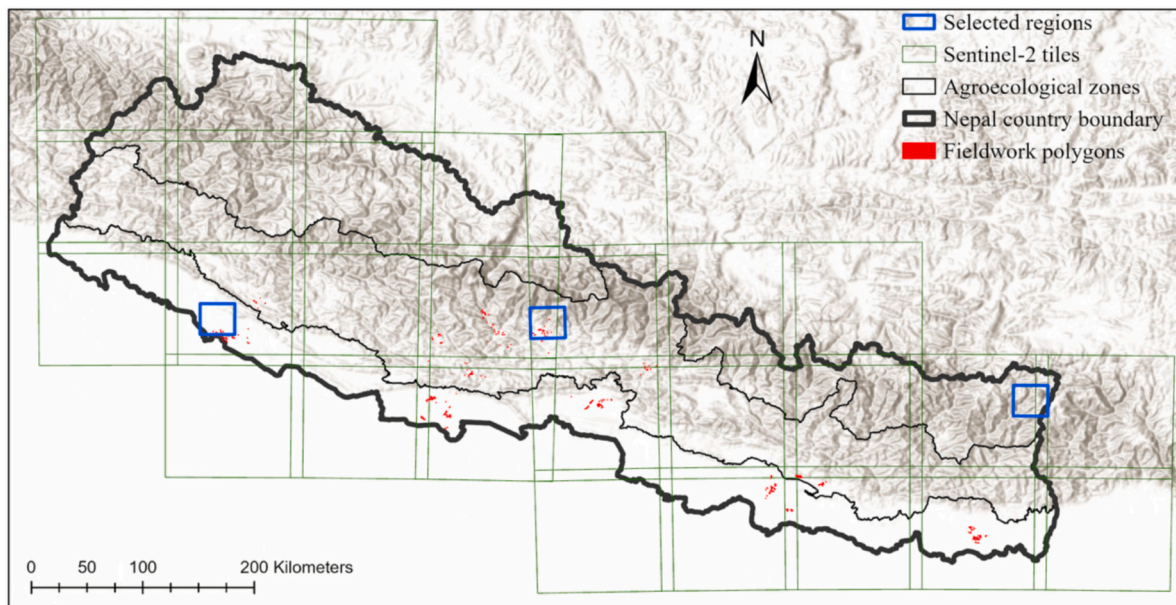


Fig. 5. Geographic location of fieldwork polygons (red color) in 2023 and they are randomly distributed at Terai zone and Hill zone. Their geographic locations intersect with 7 tiles. From left to right: 44RNS, 44RQS, 44RQR, 45RTL, 45RUK, 45RVK, 45RWK. The three focal areas with 2750 by 2750 pixels (blue boxes) are in 44RNS, 45RTM, and 45RWL from left to right.

90%, with an average value of 91.39%. The only exception was setup 3 in the model C, which achieved a slightly lower overall accuracy of 89.51%. Among the three models, model A consistently performed best, attaining an average overall accuracy of 93.09%. Models B and C, which focused on a single crop class, had average overall accuracies of 90.71% and 90.37%, respectively. Notably, setup 5 and 6, both leveraged multi-year training data combined with DOY positional encoding, produced higher overall accuracies, with setup 6 yielding the strongest performance across all three binary models.

Table 6 summarized class-specific performance across all setups and a 3-class model trained with all three years reference data. Crop classes, including all crops in model A, other crops in model B, and staple crops in model C, consistently achieved higher producer accuracy than the non-crop class, indicating stronger discrimination for crop types. Notably, setup 6 yielded the highest producer accuracy for non-crop class, as attributable to the inclusion of diverse non-crop land cover types across the 2011–2023 period, which enabled the model to better capture temporal heterogeneity. User accuracy results indicated high classification reliability across all classes of model A, B, and C, with values exceeding 88.91%. However, the staple and other crop classes showed marginally lower user accuracy by 3.82% and 3.97% compared with all crops class. Whereas non-crop class achieved an average user accuracy 92.34%, substantially higher than the associated producer accuracy (78.02%). F1-score, which balances omission and commission errors, further confirmed superior performance for crop classes relative to the non-crop class. Overall, setup 6, which leverages multi-year unsmooth time series data, consistently outperformed others.

5.2. Fieldwork validation

See Fig. 7 summarized average overall, producer and user accuracies for staple and other crops across probability thresholds using 2023 field-survey polygons. Overall accuracy for staple crops ranged from 83.73% to 93.96% and remained relatively stable across thresholds, with user accuracy consistently exceeding producer accuracy. For other crops, overall accuracy varied from 70.73% to 98.96%. Increasing thresholds led to lower producer accuracy but higher user accuracy, indicating a trade-off between omission and commission errors. Classification performance varied spatially, with tiles 45RUK and 45RVK showing comparatively lower accuracies (Fig. 7), likely due to smaller and more fragmented fields (Joshi et al., 2021).

Table 7 shows the optimal threshold combinations for 2023. Thresholds of 0.79 (staple crops) and 0.95 (other crops) achieved producer accuracies of 98.88% and 93.43%, respectively, while thresholds of 0.79 and 0.96 yielded user accuracies of 91.04% and 99.90%. These

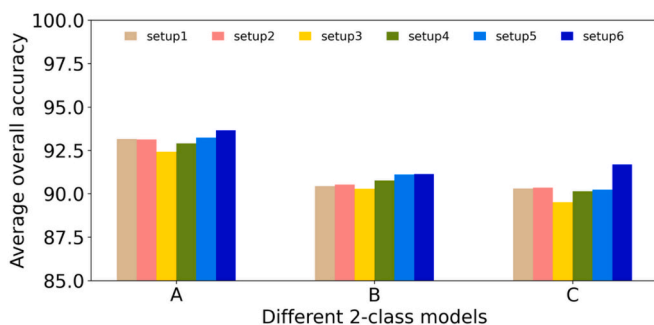


Fig. 6. Average overall accuracy for each 2-class model. Different color means different setup. The x-axis denotes the model. A: staple and other crops vs. non-crop; B: other crops vs. non-crop; C: staple crops vs. non-crop. Setups from left to right in each model: GEE data in 2022 without DOY in the model (setup 1), GEE data in 2022 with DOY in the model (setup 2), ESA data in 2022 with DOY in the model (setup 3), GEE data in 2022 with DOY in the model but without red edge bands (setup 4), GEE data in 2022 and 2023 with DOY in the model (setup 5), and GEE data from 2021 to 2023 with DOY in the model (setup 6).

combinations resulted in overall accuracies of 96.27% and 94.83%, respectively. Overall, the results supported the effectiveness of the threshold-based integration of binary models for generating a multi-class result.

5.3. Model sensitivity analysis

Model sensitivity was assessed through visual comparison between the staple crop maps and the ESRI 10 m land cover product (Fig. 8). Across three representative subregions, the maps showed strong agreement with true-color Sentinel-2 imagery and consistently captured the dominant spatial extent of staple crop cultivation.

In the Terai zone (44RNS; Fig. 8A), predicted crop distributions closely aligned with ESRI cropland, while providing finer spatial detail and explicit separation of staple crops. Minor commission errors occurred along forest-cropland boundaries, where forest edges were occasionally misclassified as staple crops. In the Hill zone (45RTM; Fig. 8B), areas predicted as crops were primarily concentrated along the Seti Gandaki and Madi Khola river corridors. Although some over-estimation occurred in *peri*-urban areas surrounding Pokhara City, the overall spatial patterns indicated robust model performance in heterogeneous landscapes. Notably, large discrepancies in the south of Pokhara City were further examined through enlarged subsets (bottom row of Fig. 8), revealing these areas were predominantly under staple crops cultivation and likely underrepresented in the ESRI product. In the Mountain zone (45RWL; Fig. 8C), the model identified localized staple crop cultivation along the Ghunsa Khola river and adjacent valley bottoms, particularly in the downstream section of the scene. These patterns aligned with documented agricultural practices in high-altitude rural regions (Rimal et al., 2018).

5.4. Qualitative and quantitative performance of national-scale maps

See Fig. 9 presented the 10 m national-scale staple crop maps from 2021 to 2023 using the thresholds showed in Table 7. Overall, the staple crops areas increased 3.91% from 2021 to 2023. Qualitatively, the spatial distribution of staple crops followed well-established agroecological gradients. Cropland was concentrated in the Terai and Hill zones, while non-crop land cover dominated the Mountain zone, consistent with previous studies (Nepal et al., 2020; Rimal et al., 2018; Uddin et al., 2015). Higher crop diversity was observed in the Terai zone, whereas staple crops dominated the Hill zone. The maps also captured localized staple crop cultivation in mountain valleys, consistent with documented maize and potato production in high-altitude regions (Rimal et al., 2018). To account for classification uncertainty arising from limited reference data, an “either” class was introduced, representing ambiguous pixels.

We further compared the mapped result (Fig. 9 left (PA)) with official statistical reports at both province and district levels for 2021 to 2023 (Fig. 10). At the province level (Fig. 10, left), the comparison yields R^2 values ranging from 0.75 to 0.83, indicating good agreement between the mapped results and statistical reports. However, the notable discrepancy was found for the Madhesh province which indicated that our result was underestimated (i.e., 33.12% lower) compared with the statistical data. The underestimation can be attributed to the region’s triple-cropping system during the year (Rimal et al., 2018), which presented challenges for crop type identification with remote sensing time series data. At the district level (Fig. 10, right), R^2 value ranged from 0.72 to 0.78, which was slightly lower than at the province level. Larger discrepancies were mainly observed in districts with extensive cropland areas, likely caused by pixels assigned to “either” class.

6. Discussion

To date, a growing number of crop type map with fine spatial resolution have become available (Ajadi et al., 2021; Li et al., 2023; Wang

Table 6

Model performance from Producer Accuracy (PA), User Accuracy (UA), and F1-score with different 2-class models and different setups. The bold values show the highest value in each model.

Data configuration	model	PA				UA				F1-score					
		all	other	staple	non-crop	all	other	staple	non-crop	all	other	staple	non-crop		
Setup 1	A	98.23			74.18	93.44			91.80	0.958			0.821		
	B			97.29			76.43		89.41			93.25		0.932	0.840
	C				95.61			81.28			89.66		91.61		0.925
Setup2	A	98.06			74.66	93.54			91.16	0.957			0.821		
	B			97.53			76.23		89.35			93.78		0.933	0.841
	C				96.14			80.53			89.34		92.47		0.926
Setup3	A	98.12			71.11	92.70			91.00	0.953			0.798		
	B			97.73			75.07		88.91			94.17		0.931	0.835
	C				95.05			80.12			89.03		90.51		0.919
Sesup4	A	98.16			73.22	93.21			91.39	0.956			0.813		
	B			98.06			75.82		89.24			95.03		0.934	0.843
	C				95.61			80.87			89.45		91.57		0.924
Setup5	A	98.04			76.07	93.61			91.56	0.958			0.831		
	B			97.21			79.71		89.95			93.87		0.934	0.862
	C				94.8			82.68			90.05		90.58		0.924
Setup6	A	97.88			79.27	94.15			91.63	0.960			0.850		
	B			96.62			81.94		89.97			93.53		0.932	0.874
	C				96.31			85.10			90.21		94.18		0.932
3-class model	/	/	73.44	37.49	81.40	/	55.82	55.53	89.05	/	0.632	0.448	0.851		

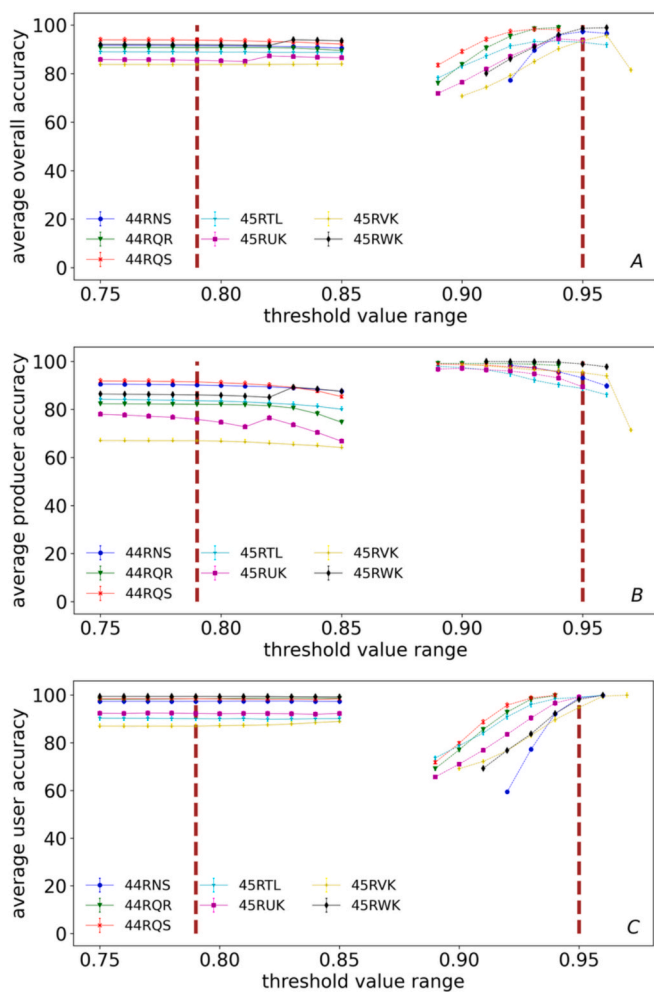


Fig. 7. The plot shows average overall accuracy (A), producer accuracy (B) and user accuracy (C) per tile. The x-axis is the range of potential threshold values for staple crops (solid lines and ranging from 0.75 to 0.86) and other crops (dotted lines ranging from 0.89 to 1.0), the y-axis is average accuracy. Different shapes in the plot area represent each tile. Vertical dashed lines indicated the selected thresholds from the average producer accuracy.

Table 7

Final paired threshold value in 2023 associated with the accuracy of staple crops and other crops compared with all fieldwork polygons. PA is producer accuracy, UA is user accuracy, and OA is overall accuracy.

2023	PA threshold	accuracy	OA	UA threshold	accuracy	OA
Staple crops	0.79	98.88%	96.27%	0.79	91.04%	94.83%
Other crops	0.95	93.43%		0.96	99.90%	

et al., 2019). However, developing countries such as Nepal have been underrepresented in crop type mapping efforts, which can be attributable to complex geographic conditions and lack of high-quality crop type reference data. Here, we used multi-year unsmooth time series Sentinel-2 L2A data with limited reference data to classify staple crops, other crops, and non-crop at 10 m resolution in Nepal for the first time.

Using deep learning models to classify crop types is increasingly prominent (Luo et al., 2022; Turkoglu et al., 2021; Xu et al., 2024). Previous studies (Luo et al., 2023; Zhang et al., 2024b) have concluded that use of an attention mechanism can improve classification performance by claiming the transformer algorithm used multi-head self-attention mechanism to better handle sequence data compared to convolutional methods from CNNs or RNNs models. The model accuracy report (Fig. 6 and Table 6) is promising, and the maps can successfully separate staple and other crops at national scale (Figs. 8 and 9). With staple crops and other crops data, day-of-year information during the growing season is critical to help interpret different crop types. To explicitly test the contribution of day-of-year on staple crop mapping, we had setup 1 and setup 2 by considering with and without day-of-year information in the model. The comparison results found that the model with day-of-year information had slightly improved overall model performance. Since crops exhibited phenological trends during growing season, the day-of-year information can relatively reflect crop growth compared with the chronological order of the data. In Nepal, crop growing seasons vary and may also include single, double or triple crop rotations (Rimal et al., 2018). Rather than constraining the analysis to a fixed seasonal window, we applied time series data throughout the year to include all possible crop growing seasons. Our results indicated that this data organization is reasonable (i.e., having ~ 0.8 R2 when comparing with official statistical reports) and aligned with findings from other studies (Belgiu and Csillik, 2018; Blickensdörfer et al., 2022;

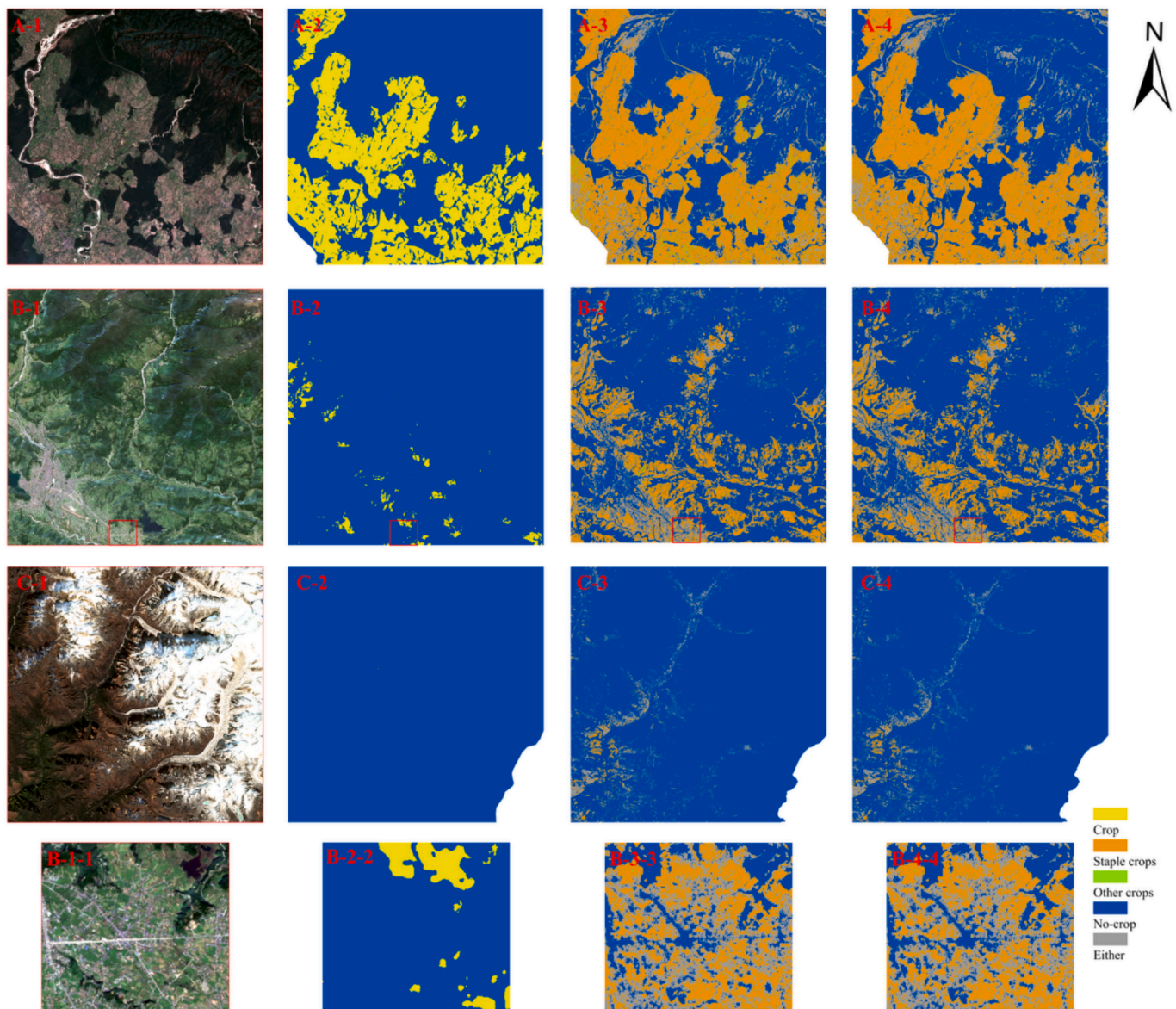


Fig. 8. The visualization of three selected regions across the country. From left to right: (1) true color images (from Sentinel-2 L2A 10 m TCI image), (2) ESRI 10 m land use and land cover map, (3) the best result from producer accuracy, and (4) the best result from user accuracy. The red box coordinate from the second row is: longitude E 84.071°, latitude N 28.165°. 44RNS (A) TCI image is on date 12/19/2023, 45RTM (B) TCI image is on date 10/22/2023, and 45RWL (C) TCI image is on date 12/03/2023. The missed corner at first and third rows are the outside of the country boundary.

Martins et al., 2020).

Time series data is essential for crop type mapping (Erinjeri et al., 2018; Gao and Zhang, 2021; Tran et al., 2022), as the crop's growing season can provide key information for crop identification from satellite images. However, with the limited reference data across multiple years, aggregating multi-year reference data has the potential to improve deep learning model performance. To test the robustness of multi-year unsmooth time series data, we tested the volume change by creating setup 2, setup 5, and setup 6. Not surprisingly, the multi-year training data had better model performance (i.e., on average higher than 0.72%) compared with the use of single year or two-year training data (Fig. 6). This underscored that deep learning models are a data-driven method where more training data can benefit the model's performance. In addition, since the model can handle any length of time series data, it doesn't matter if the image is contaminated with cloud or not. We still set the cloud cover percentage to less than 30% based on the computational cost and processing time after testing the model performance with cloud cover percentage less than 70% (i.e., model performance

showed the average overall accuracy from three models increased 0.76%). Notably, we didn't add traditional methods (i.e., composite or percentile methods) to process time series data in our setups because previous studies have concluded that irregular time series data can yield better performance (Che et al., 2024; Zhang et al., 2024a).

Although the results are promising, some limitations and uncertainties still exist. The use of smartphones for reference data has the potential to introduce bias as some locations could have poorer reception, and potential human-based bias or errors. For instance, because smartphones are mobile devices, recorded point locations may be imprecise if they are captured near field boundaries or from non-field locations such as farmers' residences. Bad reception is common in Nepal, and the company created the E-Chautaris to solve the issue, however the quality of these points could be low because of the distance between the E-Chautaris and a crop field. To mitigate these issues, instead of using all original datasets provided by the GeoKrishi mobile application, we kept 13,204 (22.94% of total points) in all three years after removing low-quality points. We posit that high-quality reference

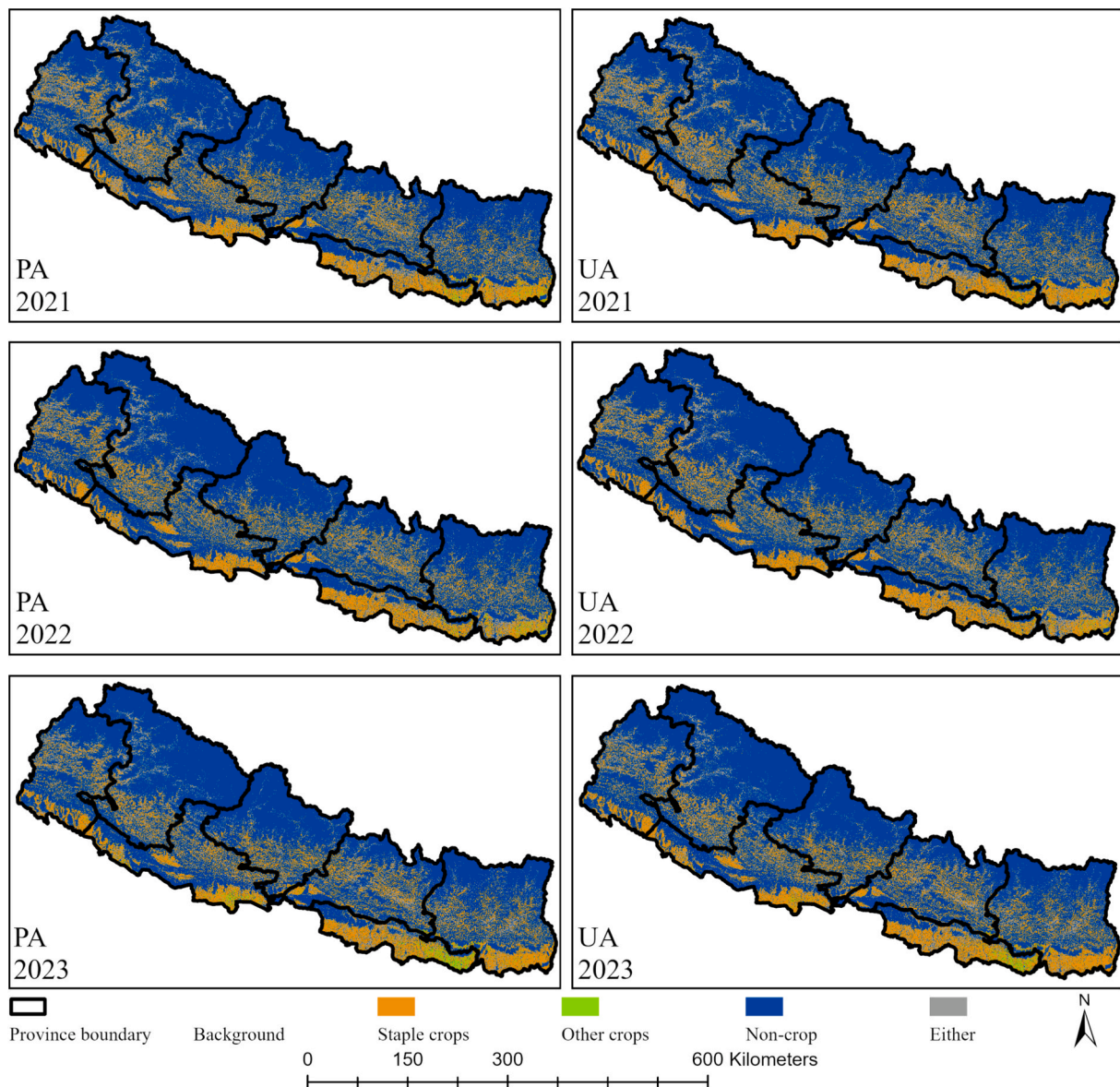


Fig. 9. Final staple crop maps at 10 m resolution in Nepal in 2021, 2022, and 2023. Left (PA) is the map with paired threshold values from the best producer accuracy. Right (UA) is the map with paired threshold values from the best user accuracy. The black lines are province boundaries, orange color is staple crop, green is other crops, blue is non-crop, white is background, and grey color is the fourth class. Top row is 2021 result, middle row is 2022 result, and bottom row is 2023 result.

data can contribute to better model overall performance, and results in Figs. 9 and 10 provide supporting evidence. Furthermore, data on individual staple crop points collected from the GeoKrishi mobile application are not enough (Table 1) to create individual classes for each crop, thus we combine them together as one grouped staple crop class in the study. Although we carefully selected reference data, the map results also showed some omission and commission errors. This limitation highlighted that quality of reference data is important and is particularly impactful in transition areas like edges of forests and mixed areas with grassland and cropland. Future work could reduce these errors by incorporating additional reference samples in such transition zones. Furthermore, integrating high-quality national-scale forest and grassland masks at 10 m spatial resolution is expected to further constrain misclassification and improve overall map accuracy.

Finally, when we applied the threshold method, since the fieldwork data crossed 7 tiles, we noticed that simply averaging all best threshold values in each tile could cause either overestimation or underestimation in certain regions. Thus, we added the “either” class (Figs. 8 and 9). The

method used in the study is compelling (Fig. 10), although the results included 12.46% pixels flagged as uncertain, underscoring the critical role of high-quality and spatially representative reference data in staple crop classification. Due to the GEE platform not being able to implement trained masked-transformer models, in this study, we classify each tile using the corresponding dataset downloaded from ESA with the same cloud mask function. Thus, some omission and commission errors could be caused by the slight inconsistency between GEE dataset and ESA dataset (Fig.A2). We anticipate that these uncertainties can be mitigated in future studies.

7. Conclusion

This study delivers the first national-scale staple crop map of Nepal from 2021 to 2023 at 10 m spatial resolution by integrating multi-year, unsmoothed Sentinel-2 L2A time series with limited reference data. By leveraging masked binary transformer models, and a probability-based thresholding strategy, we successfully produced staple crops, other

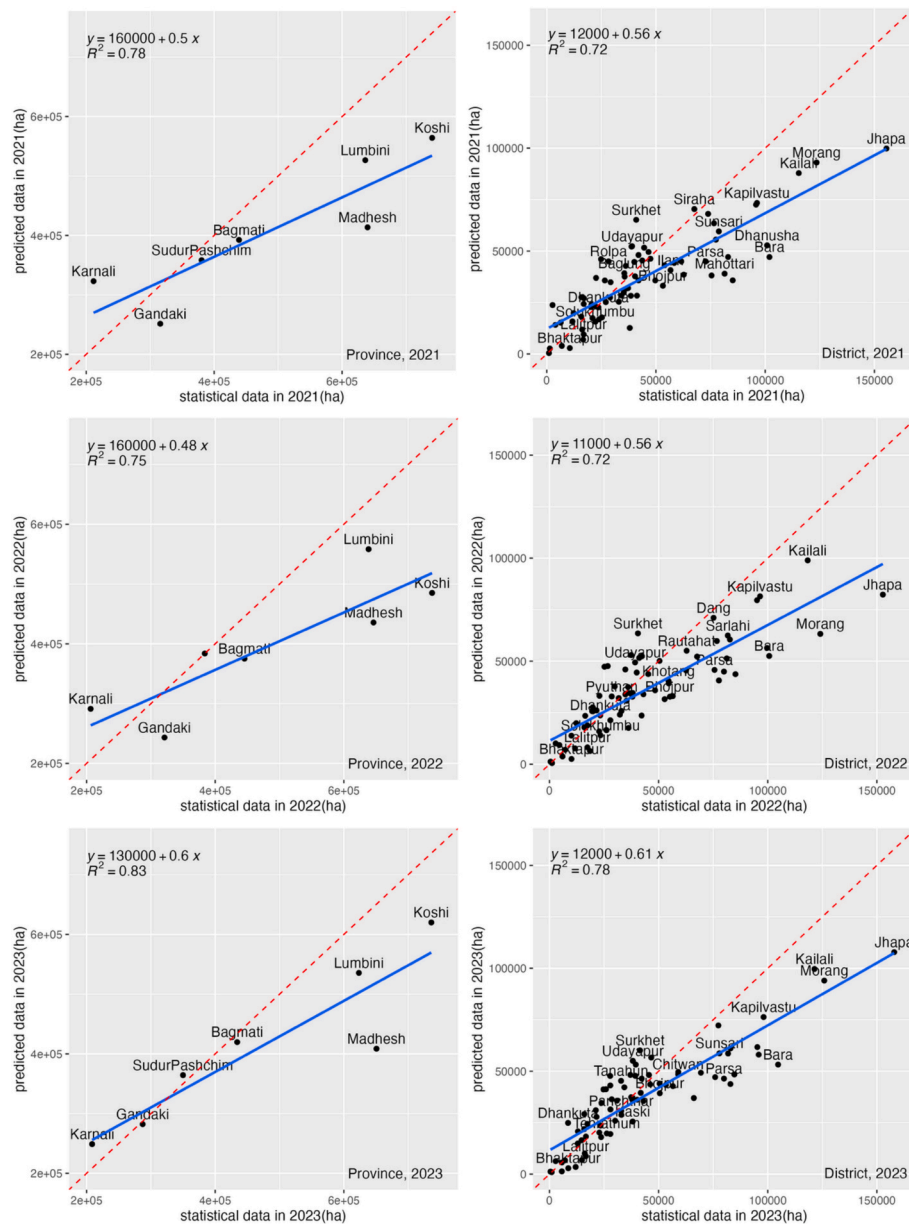


Fig. 10. Scatterplot of classified staple crop area and statistical data at province level and district level in Nepal from 2021 to 2023. The x-axis is the officially reported staple crop plantation area from the statistical data, and the y-axis is the estimated staple crop area from our result. Left is province (7) level, right is district (77) level. Blue line is the regression line between statistical data and our map result. Red dotted line is the diagonal line with slope is equal to 1 and it means deviations from this line indicates overestimate or underestimate of crop area by our result.

crops, and non-crop in data-scarce environment. Results demonstrate that models trained on multi-year unsmoothed time series with day-of-year positional encoding achieve the highest overall accuracy, effectively capturing spatial and temporal heterogeneity in staple crop distributions. The proposed probability thresholding approach successfully integrates binary classification outputs into a multi-class map, yielding high overall accuracy and strong agreement with official agricultural statistics. These findings highlight the robustness of the framework and its suitability for large-area applications. Future work may further enhance performance and efficiency through transfer learning and large foundation models, which hold promise for reducing training data demands.

The final staple crop maps at 10 m spatial resolution from 2021 to 2023 including all 28 individual tiles and national map are available at: <https://zenodo.org/records/15025615>.

CRediT authorship contribution statement

Dong Luo: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Kunwar K. Singh:** Writing – review & editing, Supervision, Project administration, Investigation, Funding acquisition, Conceptualization. **Seth Goodman:** Writing – review & editing, Methodology, Data curation. **Ariel BenYishay:** Project administration, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

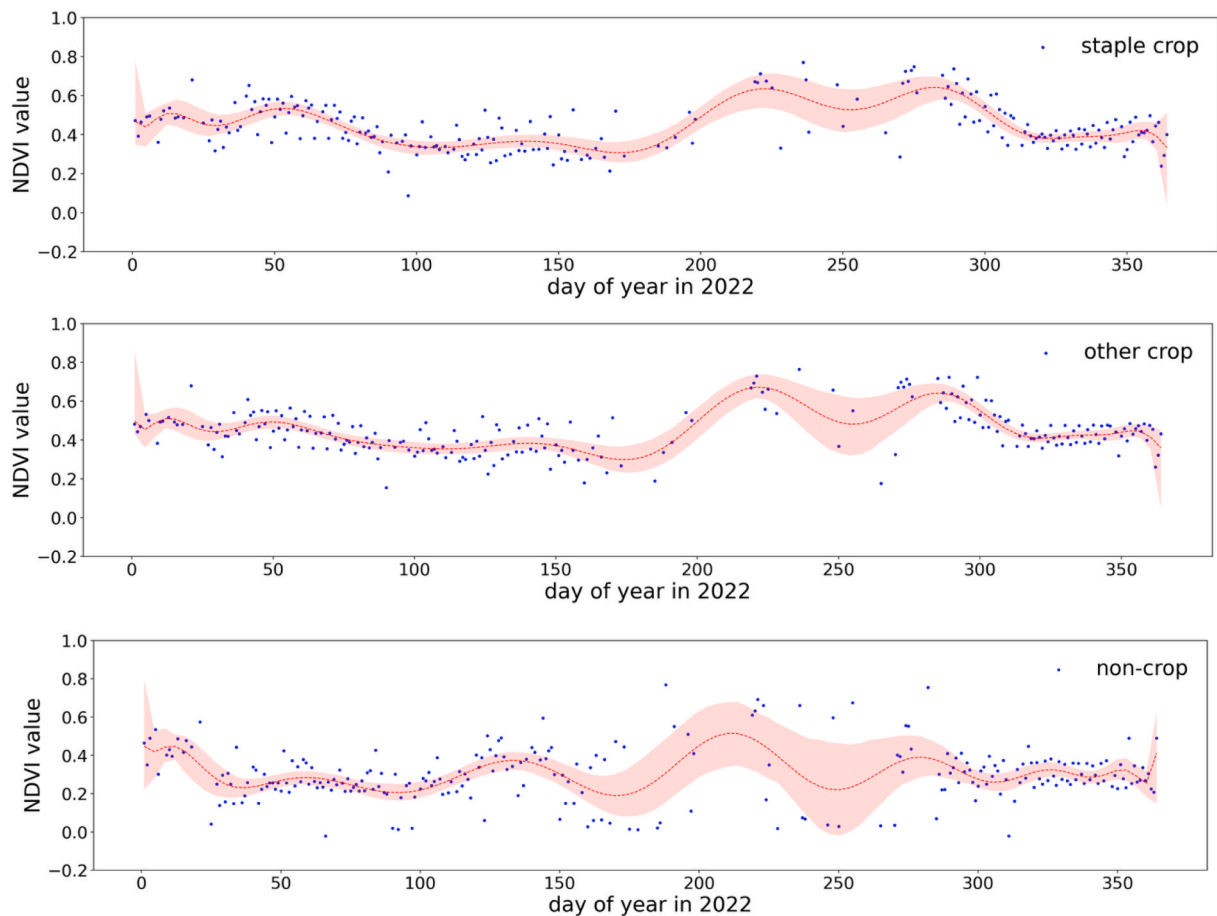


Fig. A1. Normalized Difference Vegetation Index (NDVI) time series for 2022 from January 1st to December 31st. The top panel represents staple crops, the middle panel depicts other crops, and bottom panel shows non-crops land cover types. The x-axis corresponds to the Julian day of the year, and the y-axis indicates NDVI values. Blue dots represent the mean NDVI observations across all available satellite acquisitions in 2022. The red dashed line shows the fitted NDVI curve, and the shaded red area denotes the 95% confidence interval of the fit.

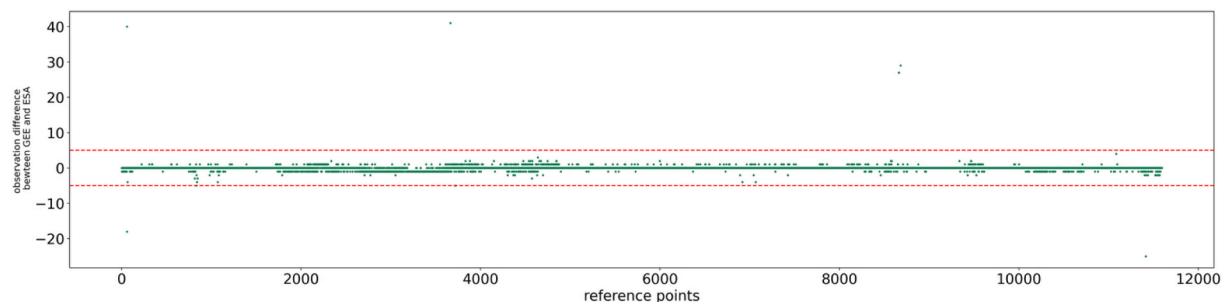


Fig. A2. The difference number of valid observations per reference point ($n = 11,594$ for 2022) between data extracted from GEE and ESA platforms. The x-axis represents individual reference points. The red dashed line range is between ± 5 . The positive red dashed line indicates that GEE have more available observations, while the negative red dashed line indicates a greater number from ESA.

Data availability

Data will be made available on request.

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